Experiment 10 : Experiment to understand array function in Data science.

Aim :

Experiment to understand array function in Data science.

Description: Understand array function using Numpy library

Program Code :

```
import numpy as np

array = np.random.randint(1, 100, 9)
print("Original Array:")
print(array)

print("\nSquare Root of Array:")
print(np.sqrt(array))

print("\nNumber of Dimensions:")
print(array.ndim)

new_array = array.reshape(3, 3)
print("\nReshaped Array:")
print(new_array)

print("\nNumber of Dimensions after Reshape:")
print(new_array.ndim)

print("\nRavel Array:")
print(new_array.ravel())

newm = new_array.reshape(3, 3)
print("\nNew Matrix:")
```

```
print(newm)

print("\nnewm[2,1:3]:")
print(newm[2, 1:3])

print("\nnewm[1:2,1:3]:")
print(newm[1:2, 1:3])

print("\nnew_array[0:3,0:0]:")
print(new_array[0:3, 0:0])

print("\nnew_array[0:2,0:1]:")
print(new_array[0:2, 0:1])

print("\nnew_array[0:3,0:1]:")
print(new_array[0:3, 0:1])

print("\nnew_array[1:3]:")
print(new_array[1:3])
```

Output :

```
Original Array:
[97 29  7 27 94  2 28 25 35]

Square Root of Array:
[9.8488578  5.38516481 2.64575131 5.19615242 9.69535971 1.41421356
 5.29150262 5.         5.91607978]

Number of Dimensions:
1
```

Reshaped Array:

[[97 29 7]

 [27 94 2]

 [28 25 35]]

Number of Dimensions after Reshape:

2

Ravel Array:

[97 29 7 27 94 2 28 25 35]

New Matrix:

[[97 29 7]

 [27 94 2]

 [28 25 35]]

newm[2,1:3]:

[25 35]

newm[1:2,1:3]:

[[94 2]]

new_array[0:3,0:0]:

[]

new_array[0:2,0:1]:

[[97]

 [27]]

new_array[0:3,0:1]:

[[97]

 [27]

 [28]]

new_array[1:3]:

[[27 94 2]

 [28 25 35]]

Result :

Thus,the we understand array function in Data science.